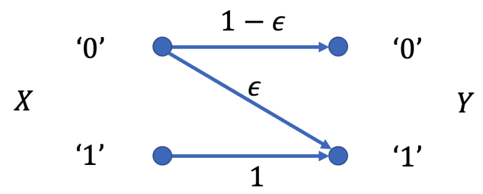
Communications and Networks (Spring 2022)

Midterm Quiz

1. (20pts) Consider a binary asymmetric channel as illustrated below:



Let p be the probability of transmitting $'0'$ and ϵ be the error probability when transmitting $'0'$.

a) Determine the joint distribution of X, Y.

b) When $p = 2/3, \epsilon = 1/2$, compute the values of $H(Y), H(Y|X = 0), H(Y|X)$.

c) Assume $\epsilon = 1/2$, derive the channel capacity $\max_p I(X;Y)$.

Solution. a)

$$P_{00} = P(X = 0, Y = 0) = p(1-\epsilon)$$
$$P_{01} = P(X = 0, Y = 1) = p\epsilon$$
$$P_{10} = P(X = 1, Y = 0) = 0$$
$$P_{11} = P(X = 1, Y = 1) = 1 - p$$

b) When $p = 2/3, \epsilon = 1/2,\ P_{00} = P_{01} = P_{11} = 1/3, P_{10} = 0$,

$$H(Y) = H(1/3) = 0.918$$
$$H(Y|X = 0) = 1$$
$$H(Y|X) = p(X = 0)H(Y|X = 0) + p(X = 1)H(Y|X = 1) = 0.667$$

c) When $\epsilon = 1/2$, the mutual information is

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y|X) \\ &= -(0.5p)\log_2(0.5p) - (1-0.5p)\log_2(1-0.5p) - p \end{aligned}$$

Differentiate w.r.t. p,

$$\frac{dI(X;Y)}{dp} = -0.5\log_2(0.5p) - \frac{0.5}{\ln 2} + 0.5\log_2(1-0.5p) + \frac{0.5}{\ln 2} - 1 = 0$$

$$\log_2 \frac{0.5p}{1-0.5p} = -2, \quad p^* = 0.4$$

Therefore, when $p^* = 0.4$, the channel capacity is

$$C = I(X;Y)|_{p=0.4} = H(0.4 * 0.5) - 0.4 = 0.322\text{bits}$$

2. (15pts) Given a signal $m(t) = \cos(800\pi t) + \cos(1600\pi t)$, we are going to encode it with the ideal sampling followed by an 8-bit uniform quantization. Suppose the signal fits exactly inside the quantization range.

   a) If $m(t)$ is sampled at 1kHz, draw the frequency spectrum of the sampled signal $M_s(f)$.

   b) Compute the quantization SNR (Signal-to-Noise Ratio) of the given signal.

Solution. a) The spectrum of the sampled signal has impulses at

$$\pm 200\text{Hz}, \pm 400\text{Hz}, \pm 600\text{Hz}, \pm 800\text{Hz}, \pm 1200\text{Hz}, \pm 1400\text{Hz}, \pm 1600\text{Hz}, \pm 1800\text{Hz}, \cdots$$

In fact the sampling rate is below the Nyquist sampling rate.

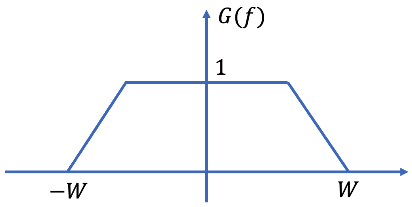
b) The average power of the signal is

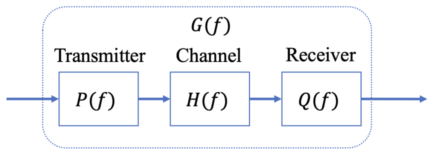
$$\begin{aligned}\sigma_m^2 &= 400 \int_0^{1/400} [\cos(800\pi t) + \cos(1600\pi t)]^2 dt \\ &= 400 \int_0^{1/400} \cos^2(800\pi t) + \cos^2(1600\pi t) + 2\cos(800\pi t)\cos(1600\pi t) dt \\ &= 1\end{aligned}$$

Thus the effective value of the signal is $D = \sigma_m / m_{\max} = 1/2$. Now we can calculate the quantization SNR by

$$10 \log_{10} \mathrm{SNR}_q = 6.02R + 20 \log_{10} D + 4.77 = 46.9\text{dB}$$

3. (25pts) The raised cosine pulse spectrum is not the only one that satisfies Nyquist Criteria. Below the trapezoid pulse spectrum $G(f)$ also satisfies this criteria, with the bandwidth $W = 20$kHz.





a) Write the symbol rate R_s for a 8PAM baseband transmission system with data rate 90kbps.

b) Given the symbol rate in a) and the zero-ISI (Inter-Symbol Interference) condition, compute and draw the pulse waveform $g(t)$.

c) A typical baseband transmission system includes a transmitter $P(f)$, a channel $H(f)$ and a matched filter $Q(f)$ at the receiver, as shown above. The channel has a transfer function of $H(f) = \exp(-|f|/f_{\text{cut}})$ that distorts the pulse, with $f_{\text{cut}} = 10$kHz. Using the trapezoid pulse spectrum in b), design and draw the frequency response of the transmitter $P(f)$ and the corresponding matched filter $Q(f)$ that would eliminate the ISI.

Solution. a) $R_s = R_b / \log_2 M = 30$kHz.

b) Let f_0 be the turning point of the spectrum, we have

$$G(f) = \begin{cases} 1, & |f| \in [0, f_0] \\ \frac{W-f}{W-f_0}, & |f| \in [f_0, W] \\ 0, & |f| \in [W, \infty] \end{cases}$$

To satisfy the zero-ISI condition, $G(f) + G(R_s - f) = C = 1$ in $f \in [0, R_s]$.

Therefore, $f_0 = R_s - W = 10$kHz and

$$G(f) = \begin{cases} 1, & |f| \in [0, 10\text{kHz}] \\ \frac{20\text{kHz}-f}{10\text{kHz}}, & |f| \in [10\text{kHz}, 20\text{kHz}] \\ 0, & |f| \in [20\text{kHz}, \infty] \end{cases} = \text{rect}\left(\frac{|f|}{10\text{kHz}}\right) * \text{rect}\left(\frac{|f|}{30\text{kHz}}\right)$$

Apply the Fourier transform and we have

$$g(t) = \text{sinc}(10\text{kHz} \times t) \cdot \text{sinc}(30\text{kHz} \times t)$$

i.e. the pulse waveform is the multiplication of two sinc functions.

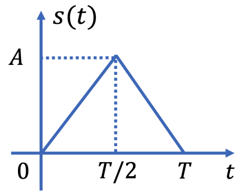
c) Since $G(f) = P(f)H(f)Q(f)$, we have

$$P(f)Q(f) = G(f)/H(f) = \begin{cases} \exp(|f|/10\text{kHz}), & |f| \in [0, 10\text{kHz}] \\ \frac{20\text{kHz}-f}{10\text{kHz}} \exp(|f|/10\text{kHz}), & |f| \in [10\text{kHz}, 20\text{kHz}] \\ 0, & |f| \in [20\text{kHz}, \infty] \end{cases}$$

Since $Q(f)$ is a matched filter of $P(f)$, one design is to take the square root,

$$P(f) = Q(f) = \sqrt{G(f)/H(f)} = \begin{cases} \exp(|f|/20\text{kHz}), & |f| \in [0, 10\text{kHz}] \\ \sqrt{\frac{20\text{kHz}-f}{10\text{kHz}}} \exp(|f|/20\text{kHz}), & |f| \in [10\text{kHz}, 20\text{kHz}] \\ 0, & |f| \in [20\text{kHz}, \infty] \end{cases}$$

4. (20pts) In a baseband transmission system, the impulse waveform $s(t)$ is given below. The double-sided PSD (Power Spectrum Density) of the noise is $n_0/2$. At the receiver, the output is sample at time T after the matched filter.



a) Determine the matched filter $h(t)$ for the receiver.

b) Determine the sampled signal output $s_o(T)$ and the sampled average noise power $\mathbb{E}[n_o^2(T)]$ after the matched filter.

c) Suppose we use a 4PAM modulation, with equiprobable symbols $a_i = \{0, 1, 2, 3\}$. If a given symbol a_i is chosen, $a_i s(t)$ is transmitted. Determine the decision rule after sampling at the receiver and derive the SEP (symbol error probability).

Solution. a) The matched filter is

$$h(t) = s(T-t) = \begin{cases} \frac{2At}{T}, & t \in [0, T/2] \\ \frac{2A(T-t)}{T}, & t \in [T/2, T] \end{cases}$$

which has the same shape as $s(t)$.

b) The sampled signal output after the matched filter is

$$s_o(T) = \int_0^T s^2(\tau)d\tau = 2\int_0^{T/2} \frac{4A^2}{T^2}\tau^2 d\tau = \frac{1}{3}A^2T$$

and the sampled average noise power is

$$\mathbb{E}[n_o^2(T)] = \int_0^T \frac{n_0}{2}|g(\tau)|^2 d\tau = \frac{1}{6}A^2Tn_0$$

c) For a 4PAM modulation with equiprobable symbols, the SEP is

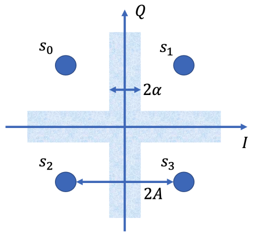
$$P_s = \frac{2(M-1)}{M}Q\left(\frac{'A'}{\sigma}\right) = \frac{3}{2}Q\left(\frac{'A'}{\sigma}\right)$$

where $'A'$ corresponds to $s_o(T)/2 = A^2T/6$ and $\sigma = \sqrt{\mathbb{E}[n_o^2(T)]} = \sqrt{A^2Tn_0/6}$.

The decision intervals for $\{0, 1, 2, 3\}$ are $\{[-\infty, \frac{A^2T}{6}], [\frac{A^2T}{6}, \frac{A^2T}{2}], [\frac{A^2T}{2}, \frac{5A^2T}{6}], [\frac{5A^2T}{6}, \infty]\}$ and the SEP is finally derived as

$$P_s = \frac{3}{2}Q\left(\frac{'A'}{\sigma}\right) = \frac{3}{2}Q\left(\sqrt{\frac{A^2T}{6n_0}}\right)$$

5. (20pts) The decision process of a QPSK transmission system is illustrated below. The distance between the symbols is $2A$ and the width of the shaded area is 2α. If the received signal falls in the shaded region, the output is an 'e', meaning an erasure, otherwise it will be decided as the nearest symbol. The average energy per bit is E_b and the double-sided PSD (Power Spectrum Density) of the noise is $n_0/2$.



a) If s_0 is transmitted, determine the probabilities that the output is s_0, s_1, s_2, s_3, and 'e'. Express your answers in $E_b, n_0, \alpha/A$.

b) If the symbols are equiprobable, what is the probability of mistaking one symbol for another? Compare your answer with the symbol error probability of the classic QPSK.

Solution. a) We treat the QPSK transmission system as 2 independent linear 2ASK systems, with $\sigma_L = \sigma_I = \sigma_Q = \sqrt{n_0/2}$. First we represent the probabilities using A, α and σ_L.

$$P_{s_0} = \left[1 - Q\left(\frac{A-\alpha}{\sigma_L}\right)\right]^2$$
$$P_{s_1} = P_{s_2} = Q\left(\frac{A+\alpha}{\sigma_L}\right)\left[1 - Q\left(\frac{A-\alpha}{\sigma_L}\right)\right]$$
$$P_{s_3} = Q^2\left(\frac{A+\alpha}{\sigma_L}\right)$$

The energy of each symbol is $E_s = 2E_b = 2A^2$, thus $A = \sqrt{E_b}$ and $\alpha = (\alpha/A)\sqrt{E_b}$. After substituting A, α, σ_L, we obtain

$$P_{s_0} = \left[1 - Q\left((1-\alpha/A)\sqrt{\frac{2E_b}{n_0}}\right)\right]^2$$
$$P_{s_1} = P_{s_2} = Q\left((1+\alpha/A)\sqrt{\frac{2E_b}{n_0}}\right)\left[1 - Q\left((1-\alpha/A)\sqrt{\frac{2E_b}{n_0}}\right)\right]$$
$$P_{s_3} = Q^2\left((1+\alpha/A)\sqrt{\frac{2E_b}{n_0}}\right)$$
$$P_e = 1 - P_{s_0} - P_{s_1} - P_{s_2} - P_{s_3} = 2Q\left((1-\alpha/A)\sqrt{\frac{2E_b}{n_0}}\right) - 2Q\left((1+\alpha/A)\sqrt{\frac{2E_b}{n_0}}\right)$$
$$- \left[Q\left((1-\alpha/A)\sqrt{\frac{2E_b}{n_0}}\right) - Q\left((1+\alpha/A)\sqrt{\frac{2E_b}{n_0}}\right)\right]^2$$

b) We directly calculate the probability of mistake,

$$\begin{aligned}
P_{\text{mistake}} &= P_{s_1} + P_{s_2} + P_{s_3} \\
&= 2Q\left((1+\alpha/A)\sqrt{\frac{2E_b}{n_0}}\right)\left[1 - Q\left((1-\alpha/A)\sqrt{\frac{2E_b}{n_0}}\right)\right] + Q^2\left((1+\alpha/A)\sqrt{\frac{2E_b}{n_0}}\right) \\
&\leq 2Q\left(\sqrt{\frac{2E_b}{n_0}}\right) + Q^2\left(\sqrt{\frac{2E_b}{n_0}}\right) = P_{\text{s,QPSK}}
\end{aligned}$$

with equality if and only if $\alpha/A = 0$.